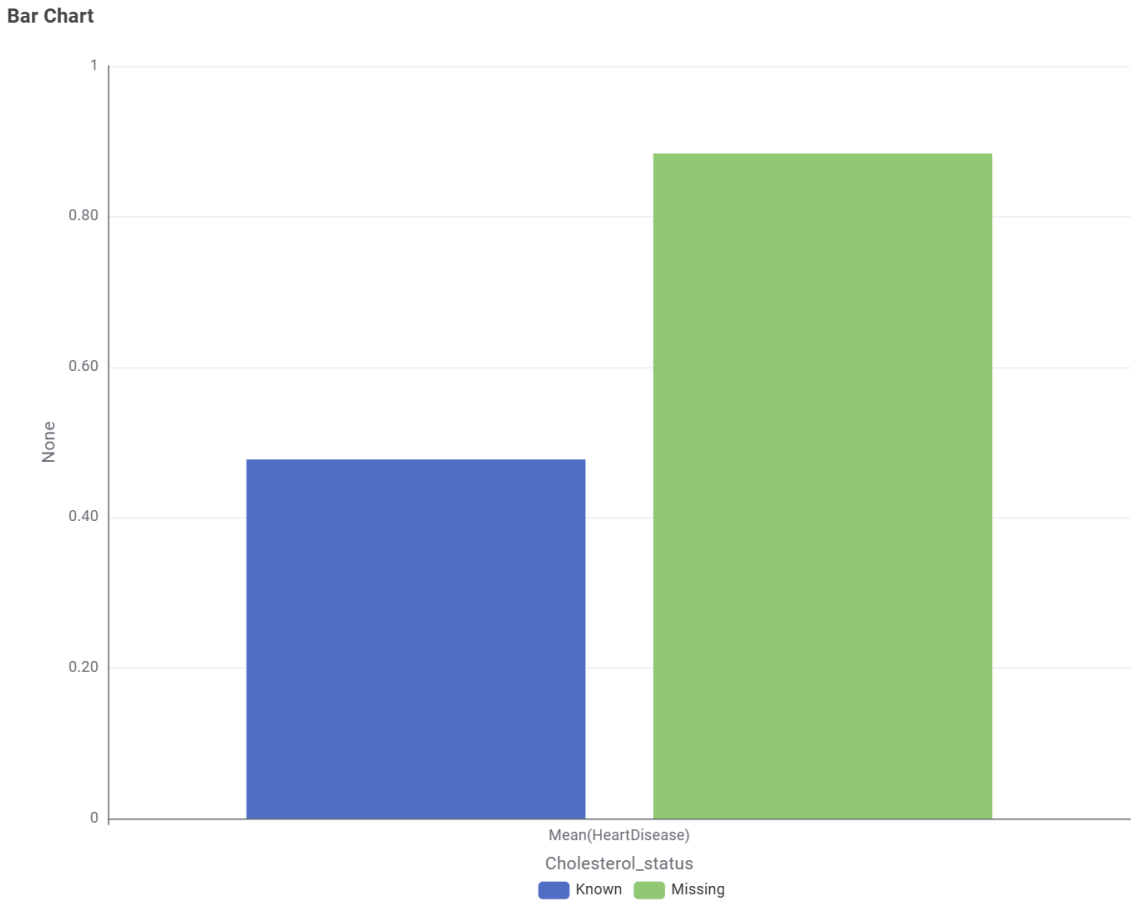
5 Question Report:

1. **The Zero-Cholesterol Profile:** Compare the 172 "Zero" patients against the rest. Are they sicker? What does the "0" likely mean clinically?

The group of 172 patients with cholesterol recorded as 0 appears significantly sicker than the rest of the dataset. Further analysis shows that 88.4% of these patients have heart disease, compared to only 47.7% in the group with valid cholesterol values. A cholesterol value of 0 is not clinically realistic, suggesting that these values represent missing or unrecorded data rather than true measurements.



2. **The Cholesterol Paradox:** Split patients into quartiles. Does higher cholesterol always equal higher disease risk in *this* specific dataset?

No, while Q4 (highest cholesterol, 276–603 mg/dL) has the highest disease rate at 57.3%, Q1 patients with the lowest cholesterol still have a 41.7% disease rate, meaning cholesterol alone is a weak and inconsistent predictor in this dataset. This is likely because many high-risk patients are already on statins suppressing their cholesterol levels, and stronger predictors like Oldpeak, ST_Slope, and ASY chest pain type are doing most of the predictive work.

View | Flow Variables

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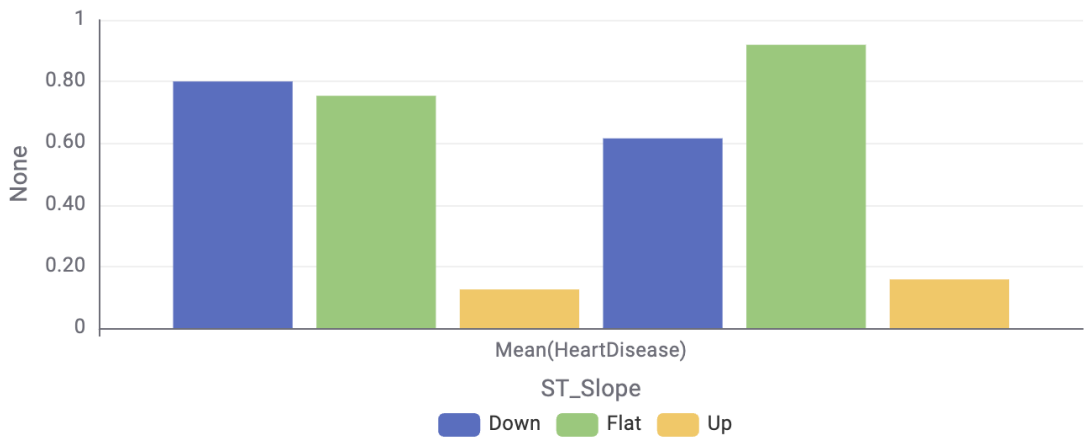

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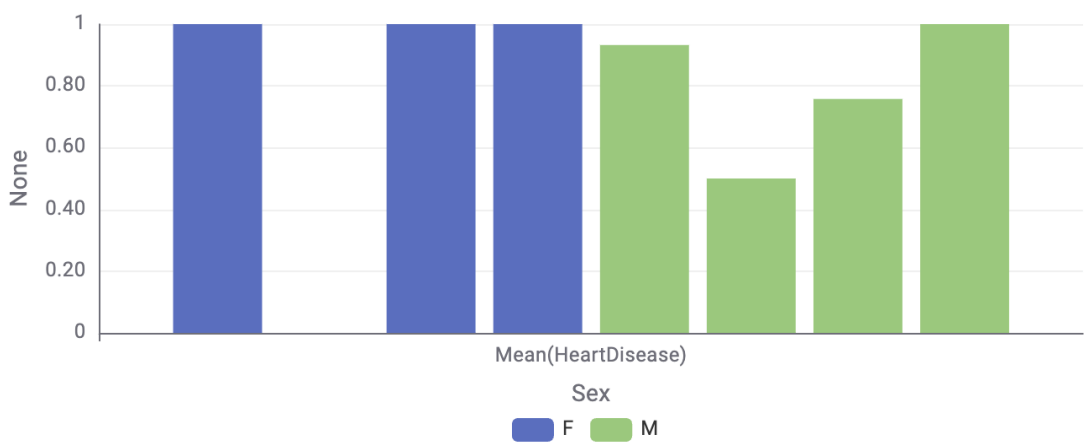


3. **The Outliers:** Identify patients under 50 with "perfect" stats (normal BP/Cholesterol/No Diabetes) who still have disease. What other features explain their diagnosis?

Included Rows (Table)

Rows: 8 | Columns: 12

#	RowID	Age (Number (...)	Sex (String)	ChestPai... (String)	RestingBP (Number (...)	Cholester... (Number (...)	FastingBS (Number (...)	RestingE... (String)	MaxHR (Number (...)	Exercise... (String)	Oldpeak (Number (...)	ST_Slope (String)	HeartDis... (Number (...)
1	Row16	38	M	ASY	110	196	0	Normal	166	N	0	Flat	1
2	Row10	46	M	ASY	118	186	0	Normal	124	N	0	Flat	1
3	Row15	38	M	ASY	110	190	0	Normal	150	Y	1	Flat	1
4	Row22	38	M	ASY	92	117	0	Normal	134	Y	2.5	Flat	1
5	Row66	49	M	NAP	118	149	0	LVH	126	N	0.8	Up	1
6	Row66	44	M	ASY	110	197	0	LVH	177	N	0	Up	1
7	Row76	40	M	ASY	110	167	0	LVH	114	Y	2	Flat	1
8	Row79	41	M	ASY	110	172	0	LVH	158	N	0	Up	1

Using the Row Filter node in KNIME, I applied conditions to isolate patients with “perfect” health metrics who were still diagnosed with heart disease. The criteria included Age < 50, RestingBP < 120, Cholesterol between 0 and 200, FastingBS = 0, and HeartDisease = 1. After applying these filters, the dataset was reduced to 8 patients.

Despite appearing healthy based on traditional risk factors, several key patterns emerged. The most notable was ChestPainType, where 7 out of the 8 patients were classified as ASY (asymptomatic), indicating they did not exhibit typical chest pain symptoms. Only one patient had NAP (non-anginal pain). This suggests that many of these individuals have silent or hidden heart disease, making it harder to detect through standard symptoms.

In addition, other features help explain their diagnosis. Most patients showed Flat ST_Slope values, which are commonly associated with abnormal heart function and reduced blood flow. There were also elevated Oldpeak values in several cases, indicating exercise-induced stress on the heart that would not be visible through resting measurements. Another notable pattern is that all patients in this filtered group are male, which aligns with known trends of higher early cardiovascular risk in men.

Overall, these findings demonstrate that even when standard indicators such as blood pressure, cholesterol, and diabetes appear normal, heart disease can still be present due to underlying electrical abnormalities (ST_Slope), stress responses (Oldpeak), and lack of visible symptoms (ASY).

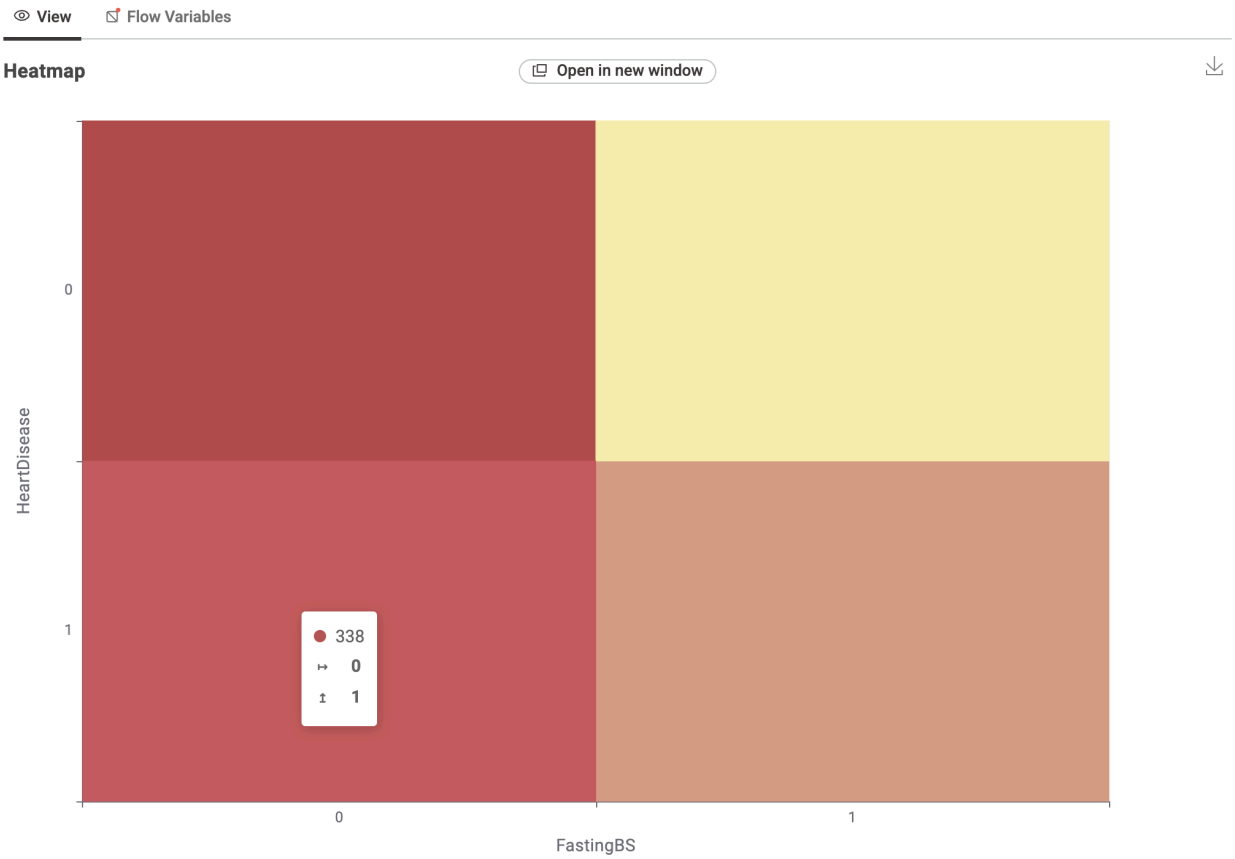
4. **Asymptomatic (ASY) Deep-Dive:** 79% of ASY patients have disease. What differentiates the 104 healthy ASY patients from the 392 sick ones?

Even though both groups are categorized as Asymptomatic (meaning they report no chest pain), the data shows they are clinically very different. Based on the Heatmap and Normalizer nodes in my workflow, the diseased group is differentiated by:

- Fasting Blood Sugar (FastingBS): My Heatmap shows that sick ASY patients are much more likely to have a value of 1 (high blood sugar). In contrast, the 104 healthy patients almost all have a 0. This suggests that high blood sugar is a major "hidden" risk factor for this group.
- ST_Slope & Oldpeak: After processing the data through the Category to Number node, there is a clear trend: the 392 sick patients typically have a Flat or Downsloping ST_Slope, while the healthy patients have an Upsloping one.

- Exercise Angina: Many of the sick patients show a "Y" for ExerciseAngina. This means that even if they feel fine at rest, their bodies show signs of distress during physical activity.

Conclusion: The main difference is that the 104 healthy patients have normal "hidden" metrics, while the 392 sick patients have normal test results (High FastingBS and Flat ST_Slopes) that only show up through clinical data, not through how the patient feels.



5. **Feature Engineering:** Calculate the **HR Reserve Ratio** (MaxHR / (220 - Age)). Does this engineered feature correlate more strongly with disease than raw $MaxHR$?